DISK DEMON 2026 Rework

Assembly tips for PCB Rev 4

DISCLAIMER
I HAVE USED THIS BOARD EXTENSIVELY WITH VARIOUS 1541 DRIVES AND C64 COMPUTERS WITHOUT ANY ISSUES. BUT, I CANNOT GUARANTEE THAT IT WILL NOT CAUSE PROBLEMS TO ANY OTHER HARDWARE OR SOFTWARE.
I ACCEPT NO RESPONSIBILITY FOR ANYTHING THAT HAPPENS AS A RESULT OF INSTALLING AND USING THE BOARD.

This rework is my tribute to the original Datel Disk Demon. Although the appearance of the new board is similar to the original design, it is not identical.

The schematic for the board is virtually the same as the original. Apart from adding some bypass capacitors, the most significant change is the replacement of the original 82S123 Bi-polar PROM with an ATF16V8 PLD (aka 'PLD').

I have also replaced the 8K2, 1K2 & 56pF resistor/capacitor combination (used on the original Disk Demon board but different to the Professional DOS values) with a single 10K resistor & 56pF capacitor (or 5K6 & 100pF). This gives an on-state period of around 328nS compared to around 336nS measured on the original Datel Disk Demon board.

Finding reliable, blank 82S123 chips is challenging in 2026. Once you do find one, it still has to be programmed which is also difficult. Additionally, some of the 82S123 equivalents do not work as expected in the Disk Demon board.

On the other hand, the ATF16V8 is still being manufactured in 2026 and available new from distributors such as Mouser, Digikey & RS. It can also be programmed with the popular XGeku T48 programmer.

The original Datel Disk Demon had both the LS123 (U2) and 82S123 PROM (U1) mounted within the footprint of the CPU socket. Although this is a compact solution, I have a 'thing' about soldering reprogrammable devices directly onto PCBs. So, I have moved U1 (ATF16V8) to an exposed part of the board allowing it to be socketed and, if necessary, easily removed &/or re-programmed.

This has increased the length of the board by about 9mm which is not an issue in the 1541. Other than the change to U1, I have kept the chip numbering the same as used by Wolfgang Moser in his excellent Professional DOS documentation.

The aim of this document is to provide some tips on how to go about making your own Disk Demon 2026 Rework using the Rev 4 PCB. It does not address making a parallel cable adaptor nor the 64 Kernal ROM adaptor.

Any comments or feedback, please post at
https://www.lemon64.com/forum/viewtopic.php?t=82454

Getting started

If your desoldering skills are up to it then I would **STRONGLY** recommend removing the original Commodore single wipe socket and replacing it with a dual wipe socket (I prefer these to the machine pin sockets). Ideally do the same for the 6522 VIA socket.

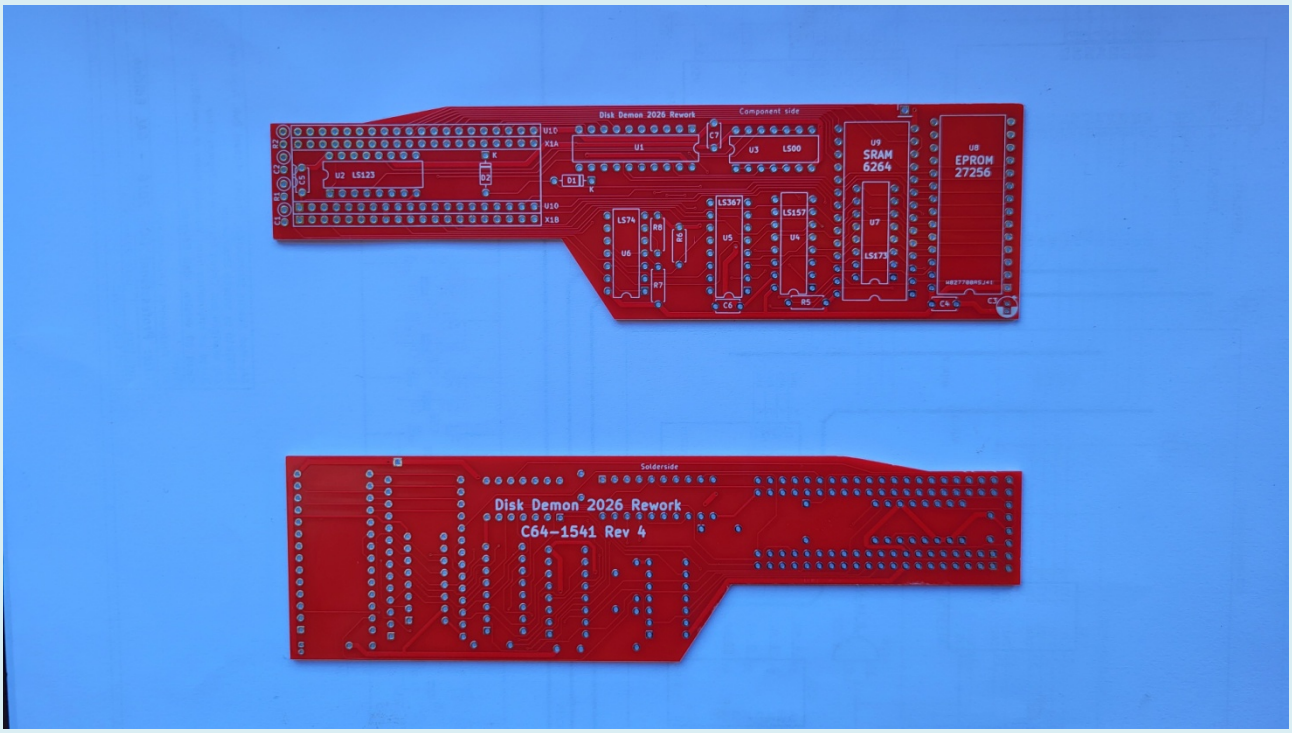
All of the components are 'Through Hole' so most of the assembly is straight forward. However, some care needs to be taken when putting in the CPU pin headers and female headers/ 40 pin socket. It is important to get the pins straight so that they are a good fit for both the 1541 system board CPU socket and the CPU itself.

Clean both sides of the PCB with Isopropyl Alcohol. It may also help to wash them in soapy water too but make sure it is completely dry before you start.

Test all components before installing them. U1 - U9 can all be at least partially tested with a T48 Programmer.

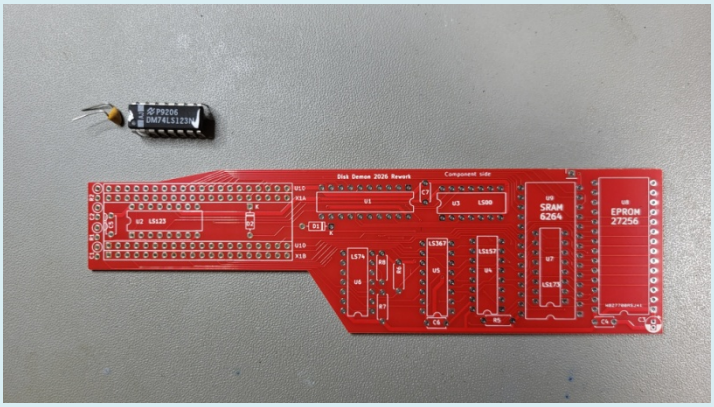
The tolerance on most of the components is not very important **APART** from C1, C2, R1 & R2. Ideally, R1 should have a value within 5% of R2 and C1 should have a value within 5% of C2 (ie matched pairs).

I normally test ten or so capacitors and choose the two that are closest both to each other in measured value and to the specified value of 100pF. Do the same for R1 & R2.



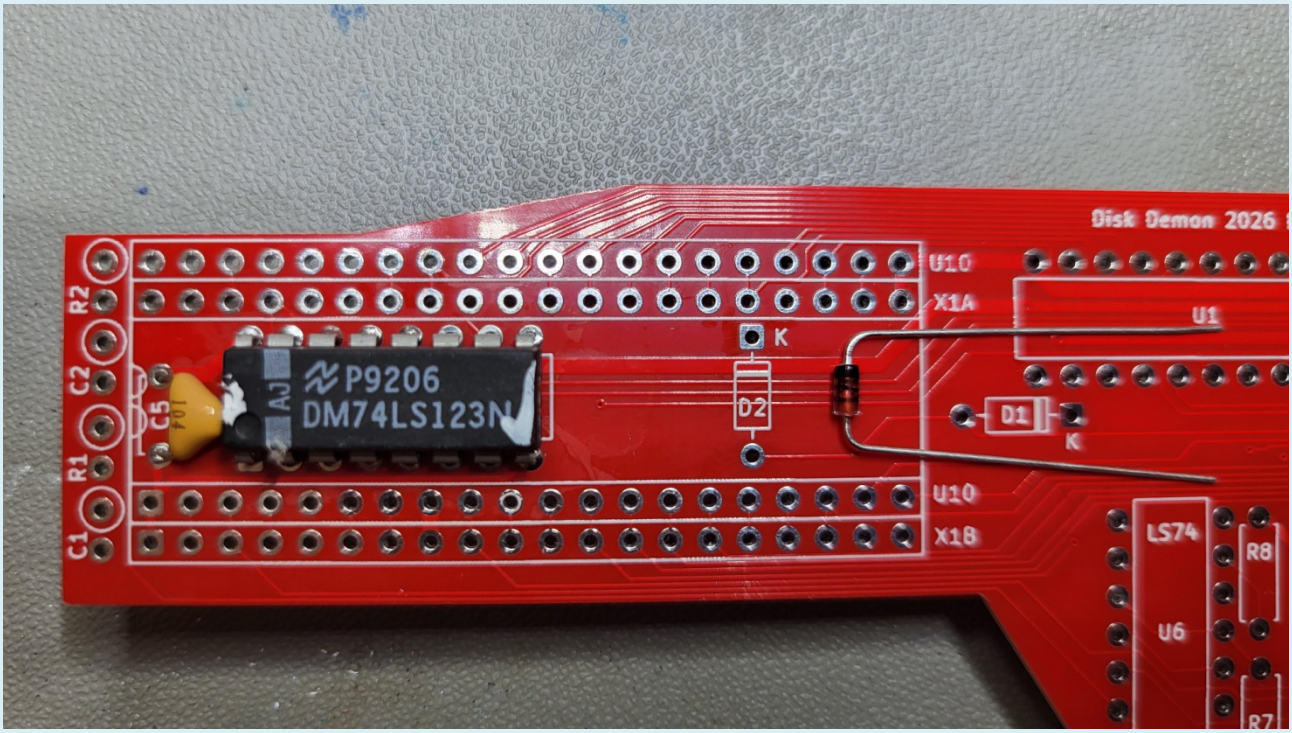
Step 1 – Install U2 and C5 (100nF)

Insert C5 and U2 into the PCB and solder into place. Leave a gap of about 1mm on the leads of C5 and bend them to a right angle. This will allow it to be folded flat next to U2



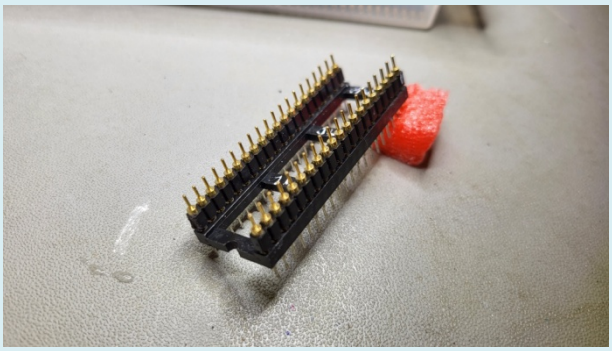
Step 2 – Install D2 (1N4148)

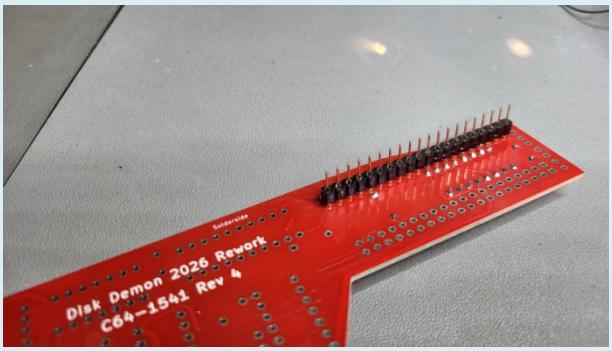
IMPORTANT. Diodes are polarity dependent. It is important that the Cathode on the diode (indicated by a black band on the diode body) is connected to the K mark on the PCB



Step 3 - Install the first CPU male pin header in X1A

In order to make sure that the pins are in exactly the right position, I normally put both rows of pin headers into the board and then plug that into a 40 pin socket. That will keep the pins straight and correctly spaced while soldering the first header.

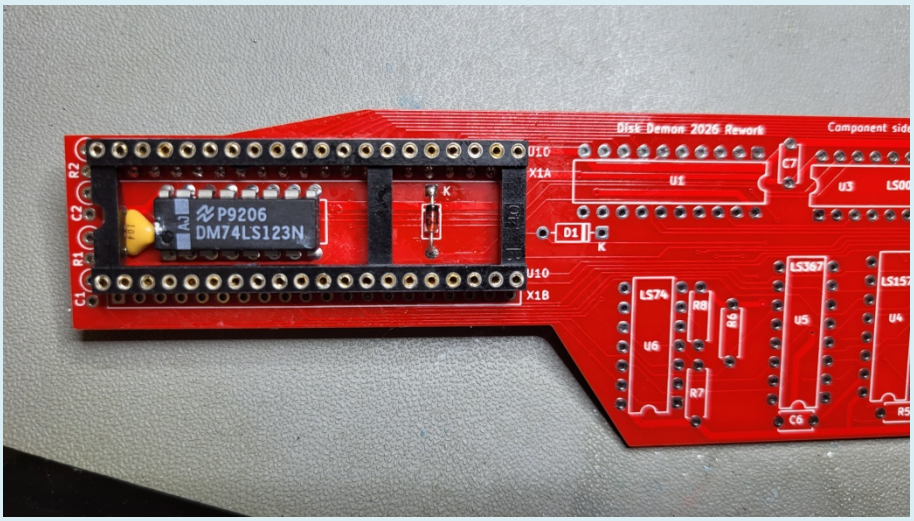




The header is inserted on the solder side of the PCB but soldered on the component side

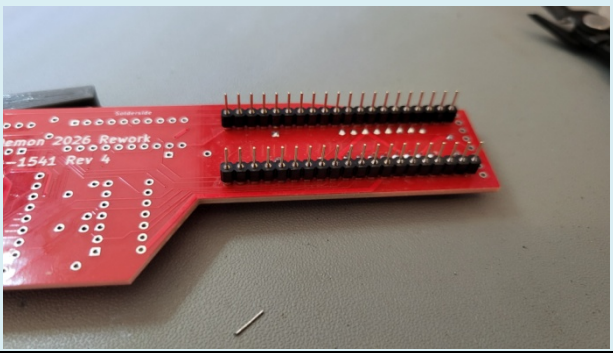
Step 4 – Install the two 20 pin female headers or 40 pin socket in U10

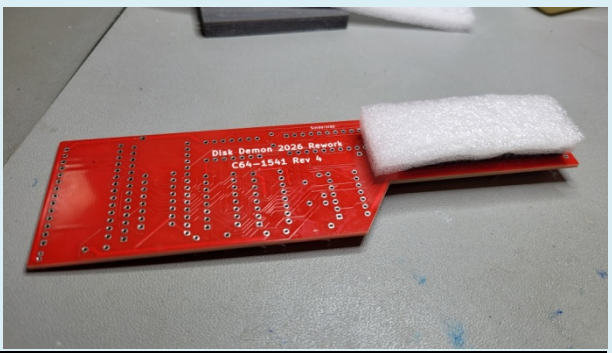
If you can find one that fits, install a 40 pin DIL machine pin socket into the U10 holes.
If you can't find one that fits, use two 20 pin female headers. Again, it may help to plug a 40 pin round pin socket into the headers to keep them straight while soldering.



Step 5 Install the second CPU (male) pin header in X1B

Same as Step 3. Once both headers are soldered in place, I recommend protecting the pins with some sort of polystyrene (or similar) foam.

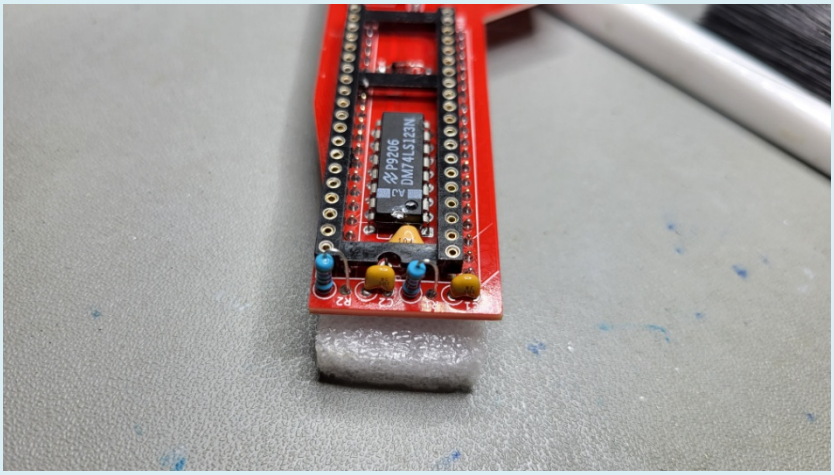




Step 6 – Install C1, C2, R1, R2 & D1 (1N4148)

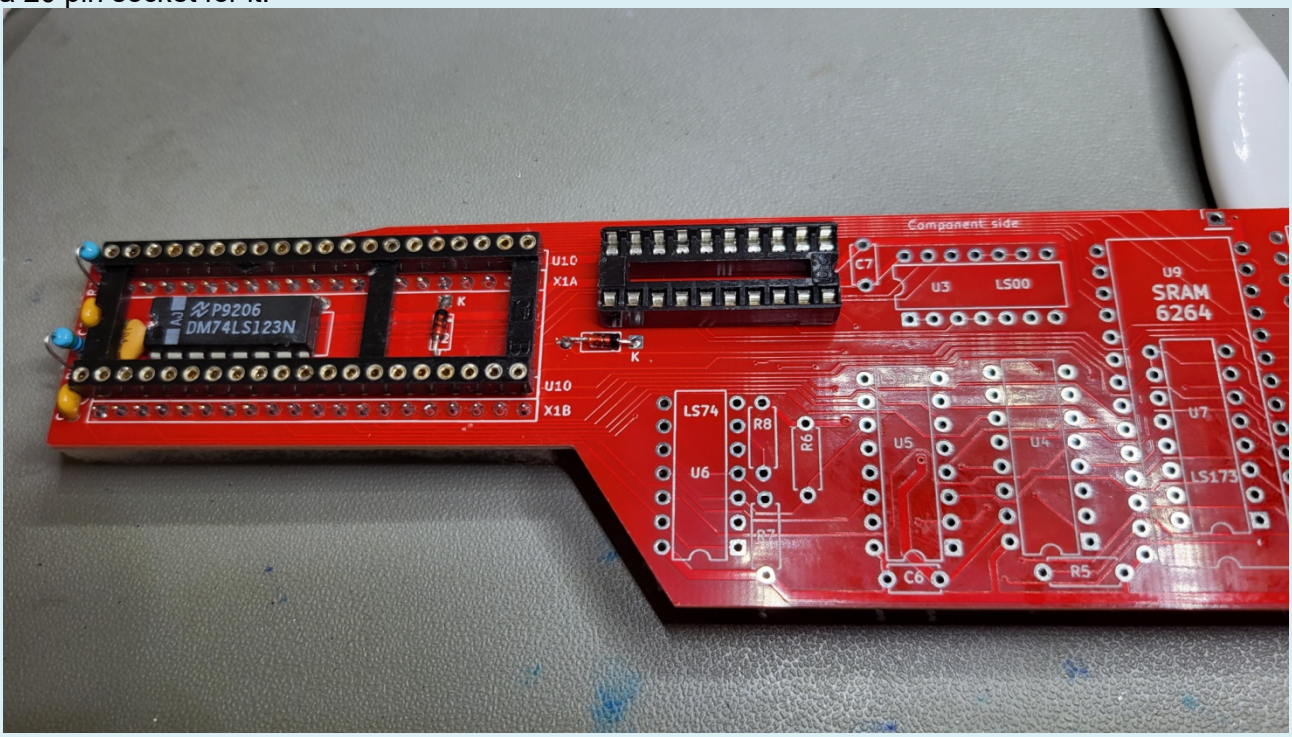
Note that R1 & R2 are mounted vertically

IMPORTANT. Diodes are polarity dependent. It is important that the Cathode on the diode (indicated by a black band on the diode body) is connected to the K mark on the PCB



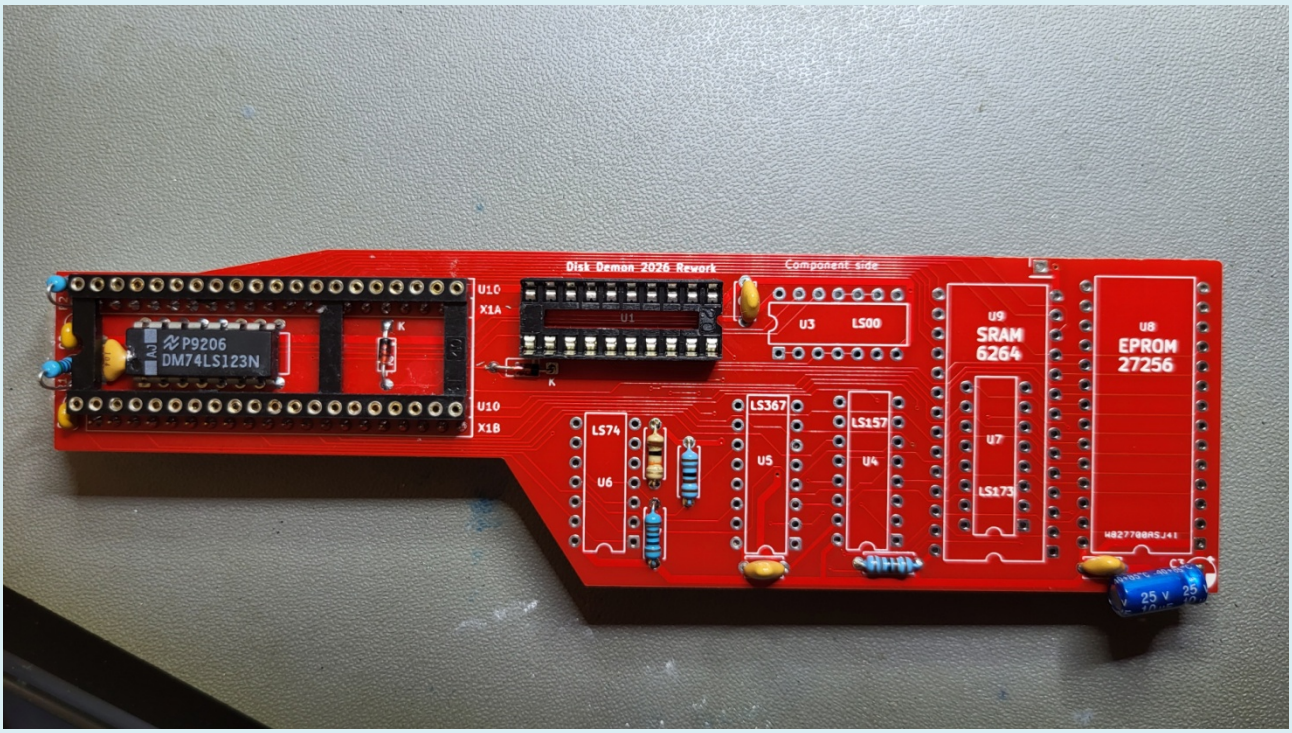
Step 7 – Install 20 pin socket for U1 (optional)

U1 is a reprogrammable 20 pin DIL IC. Unless you plan to solder U1 directly onto the PCB, install a 20 pin socket for it.



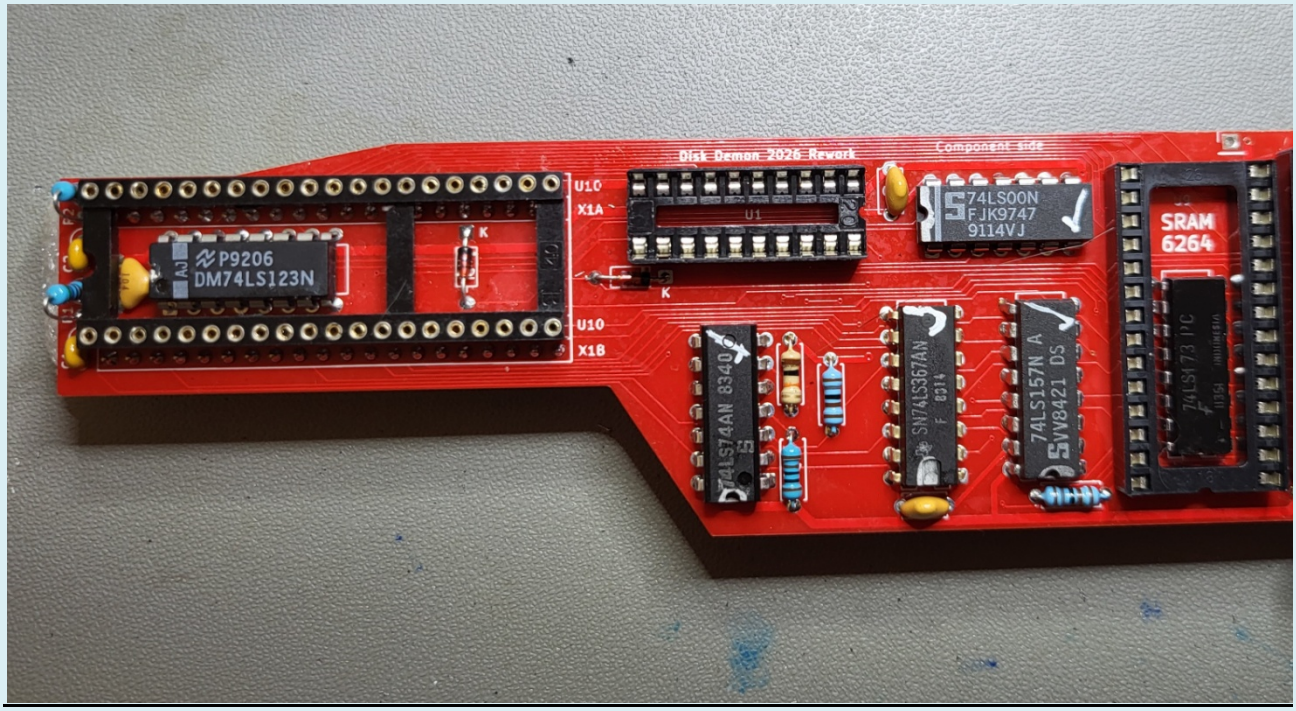
Step 8 – Install R5 (1K8), R6(1K8), R7 (1K0), R8(10K), C3 (10uF), C4, C6 & C7 (all 100nF)

It is a good idea to leave 1-2mm gap between the PCB and the body of C3 so it can be folded over if needed. Remember to check the polarity of C3.



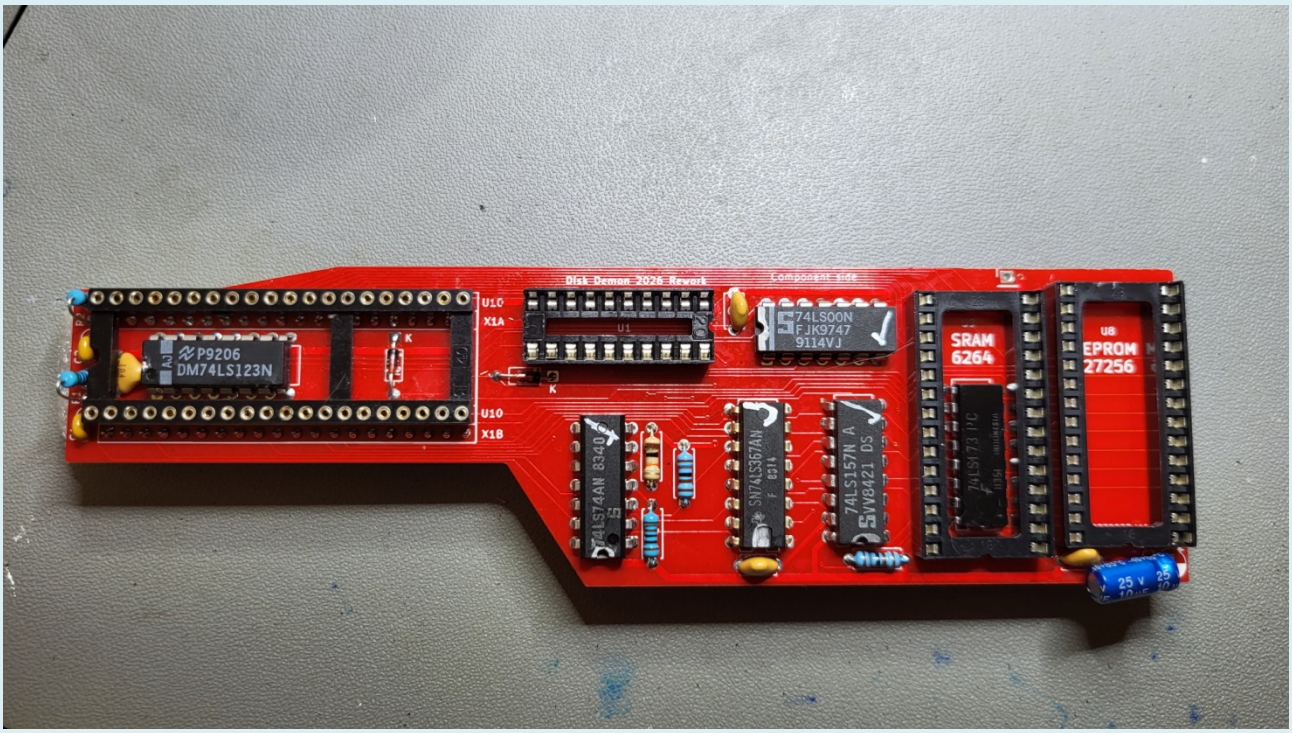
Step 9 – Install U3, U4, U5, U6 & U7

U7 must be soldered directly onto the PCB. U3, U4, U5, U6 can be socketed or soldered directly onto the PCB



Step 10 – Install the 28 pin sockets

Depending on the type of socket used, the socket for **U9** may need to have a centre cross bar removed so that it can fit over **U7**. I just use a pair of wire cutters. Once that is done, both sockets can be soldered into place.



Soldering complete !

Next steps: Program the PLD & EPROM and insert them, along with the RAM, into their sockets….

Program the PLD (U1) & install

U1, the ATF16V8 PLD, needs to be programmed before it can be used. This can be done easily with a XGeku T48 programmer. I have been testing the **ATF16V8B-15PU** and the **ATF16V8C-7PU.** Both work without any issues using the **DD-ATF16V8B.jed** JEDEC file

Use the file **DD-ATF16V8B.jed**
Deselect the ***'Encrypt Chip'*** and ***'Lock BIT'*** options. It may be ok to use these options but I have not tested that.

Insert the programmed PLD into the **U1** socket.

***** Note that it faces the opposite way to U10 and U3 *****

Program the EPROM (U8) & install

U8 (27256 EPROM) needs to be programmed before it can be used. This can be done easily with a XGeku T48 programmer.

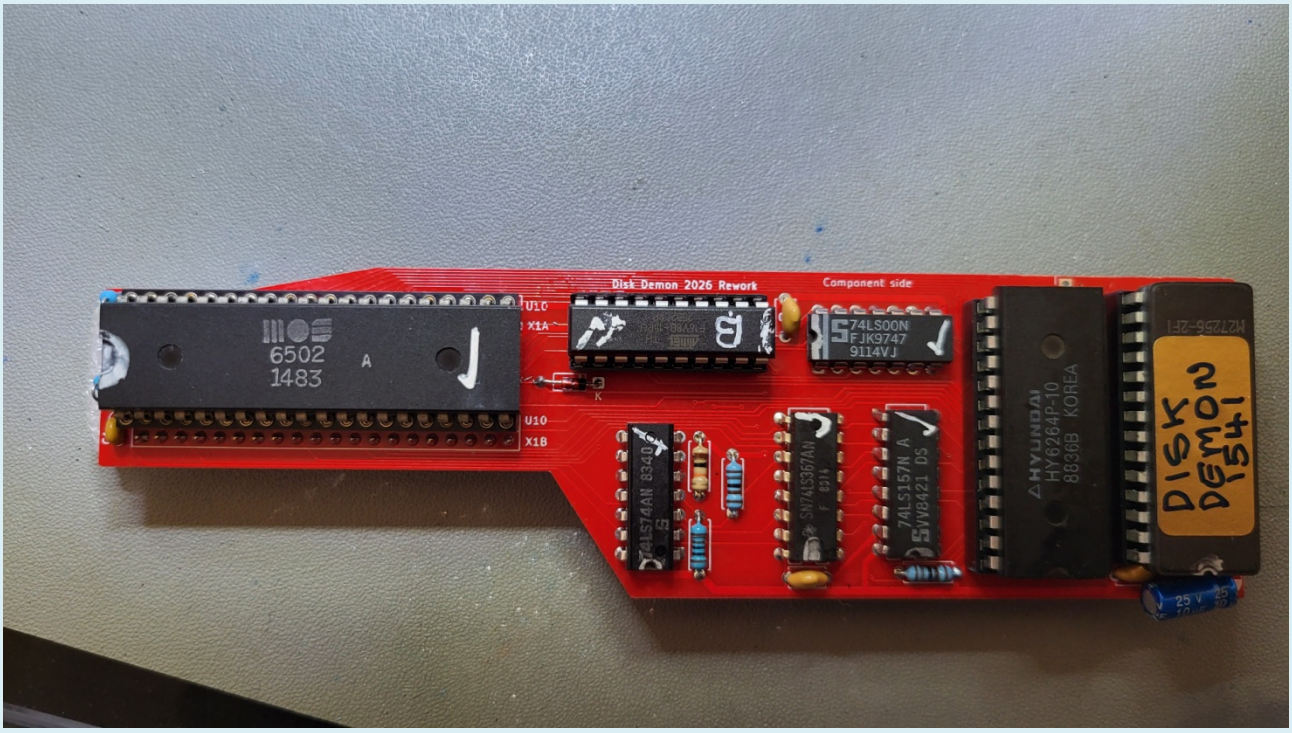
Use the file **DISKDEMON_Original_1541_ROM.bin**

Alternatively, use a OneROM Fire 28. This is a modern day substitute for the EPROM. It can be programmed just using a web browser through a USB cable attached to a computer

Once done, insert the Fire 28 or EPROM into the **U8** socket (marked **EPROM** on the PCB)

Install the RAM (U9)

Finally, the 8k SRAM chip **U9** (Eg Toshiba 5565 or Hitachi 6264), should be inserted into the **U9** socket (marked **SRAM** on the PCB)



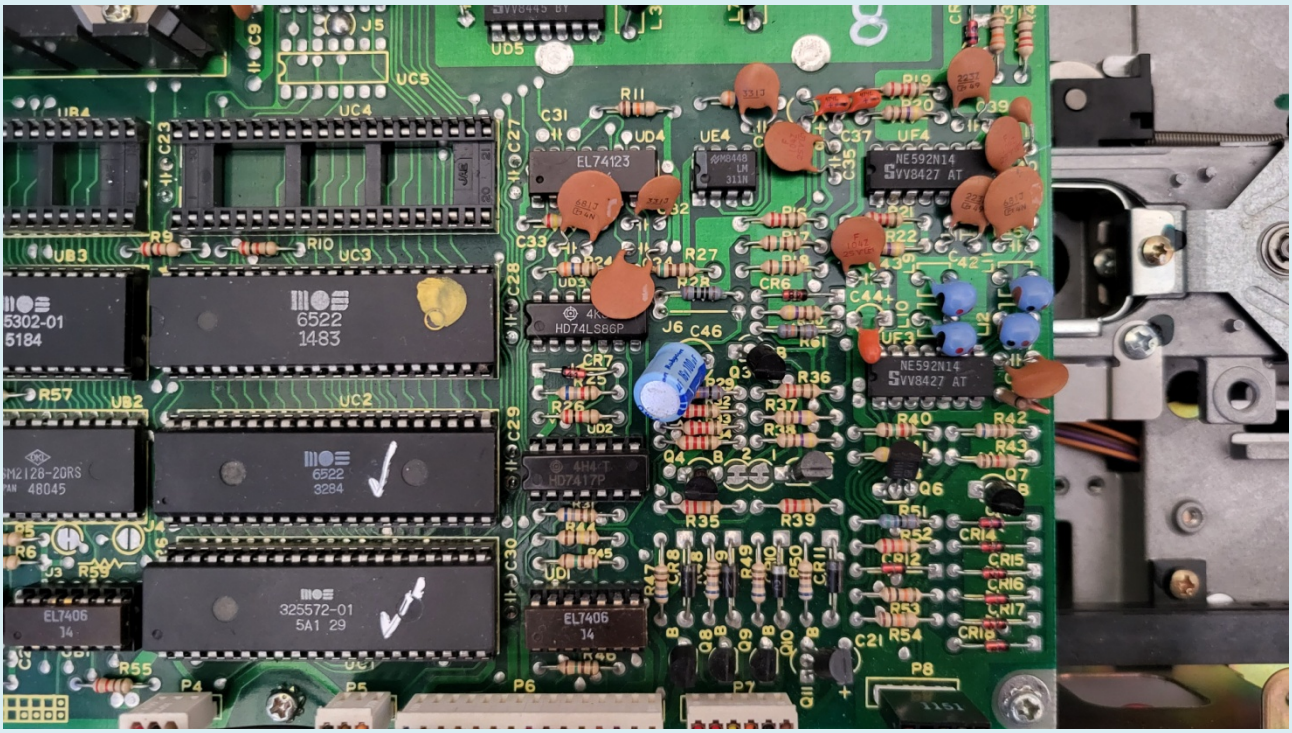
Final Steps

Remove the ROM chip from the UB4 socket on the 1541 System Board

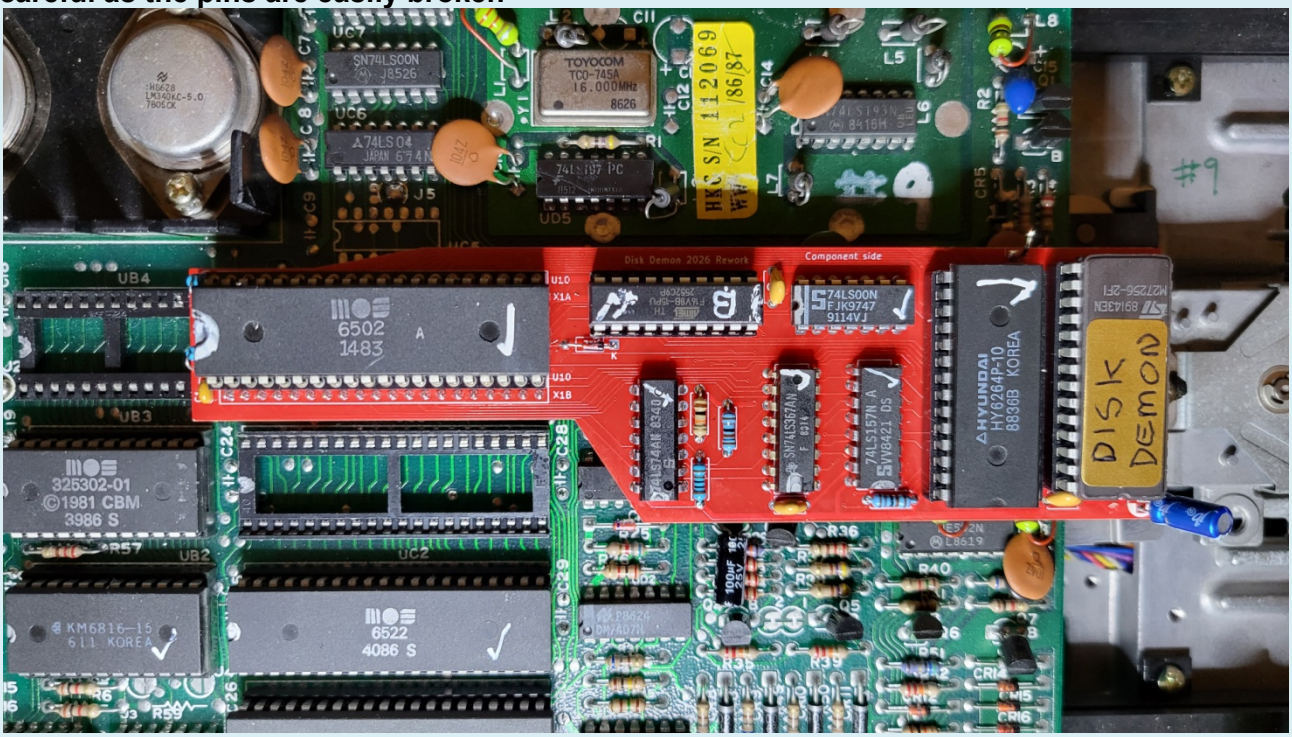
Remove the 6502 processor from the **UC4** socket on the 1541 System Board and insert it into the CPU socket (or female headers) on the Disk Demon 2026 Rework PCB.

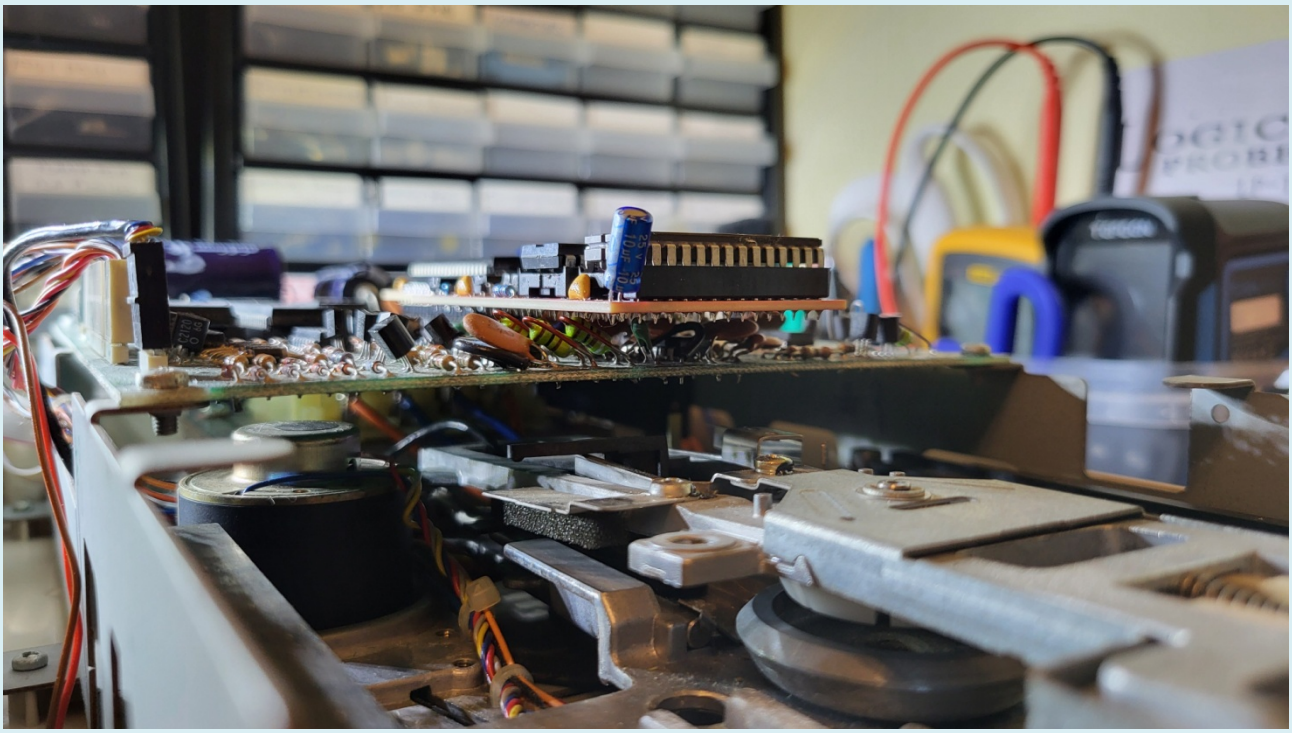
Before plugging in the Disk Demon 2026 Rework PCB, check the components on 1541 system board that will be underneath the board after it has been installed.

If there are any components that are likely to come into contact with the bottom of the PCB then gently push them down to allow a clearance. Pay particular attention to the electrolytic capacitor, C46



Carefully plug the Disk Demon 2026 Rework PCB into the vacant 6502 socket (UC4). Be very careful as the pins are easily broken





Cross your fingers and turn on the 1541 disk drive.

The green light should come on and stay on as usual.

The red light should flash on for less than 1 second and then go off. If it stays on for longer than 1 second then switch off immediately & check everything.

The next step is to install the parallel adaptor into the 6522 socket and connect it to the C64 User Port. It is a tight fit but I have tried it with several different popular parallel cables and there is room.

After that, install the Disk Demon Kernal ROM in the C64 and check the C64 startup screen

Common Installation Problems

Red data light stays on when drive is switched on

- Check the Disk Demon board is correctly inserted in the 1541 socket (UC4)
- Check the CPU is seated correctly in the Disk Demon board socket.
- One of the ICs is in backwards or bad solder joint

Try removing and replacing both the PCB & the CPU. This might be a good time to spray in some contact cleaner.

Check all the ICs are right way round and look for bad joints

Incomplete message after status command '@' or '@?' and board hangs

Instead of responding with the normal message, the C64 displays something like
'73,C', '3' or '73, CB' and then hangs.

This can happen if the original ROM in UC4 has not been removed.

There is also a very rare fault that so far has only occurred once. It was caused by the SRAM chip (U9). If possible, try a different SRAM chip.

If that is not practical, replacing the EPROM (U8) (ie not the SRAM) with a OneROM may resolve the problem.

Failing that, it is necessary to cut the topside track between U6, pin 6 and and R8. This will disable the **'@D0'** command but it should sort out the issue.

